Green Home Score & Retrofit Engine

How our system works, in plain English, and who builds what

What we are building

A web app where a homeowner can see how green their home is and what to do about it. It answers four questions:

- **How green is my home today?** A score from 0 to 100, with the reasons behind it.
- **What should I improve?** A short, ordered list of upgrades that suit this home.
- **How much energy and carbon could I save?** Electricity and gas shown separately.
- **Could my house be worth more?** A rough range, clearly marked as an estimate.

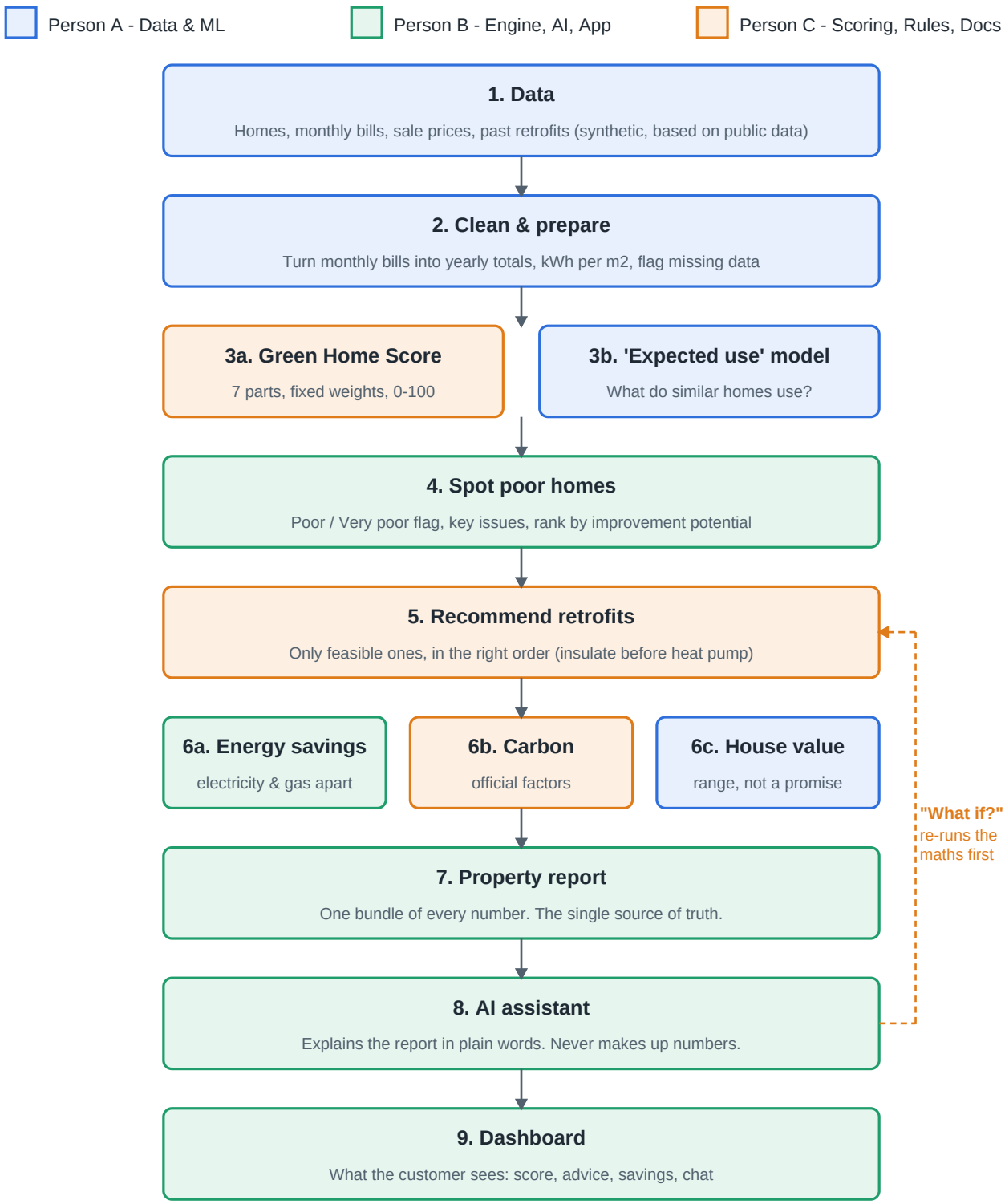
The five golden rules

The judges check these closely. Breaking any one of them costs marks.

#	Rule	What it means
1	**The AI never does the maths.**	Every number (score, savings, carbon, value) comes from our own code. The AI chat only explains those numbers. For a 'what if' question, the code reruns first and the AI explains the new result.
2	**Electricity and gas are always separate.**	A heat pump cuts gas but adds electricity. We must show both, not only the total.
3	**Every number has a source.**	Retrofit effects, costs and carbon factors all sit in config files with a note of where they came from.
4	**Be honest about uncertainty.**	Savings are 'estimates'. House value is an 'indicative range'. Missing data says 'unavailable', never 0.
5	**No cheating in testing.**	A home (and its owner) appears in only one of train / validation / test. Models never see information from the future.

How it flows

Data goes in at the top. The customer sees the dashboard at the bottom. Box colours show who builds each part.



Each step in plain words

1. The data

We cannot use real customer data, so we generate a realistic fake town of about 10,000-20,000 homes. The numbers are tuned to real public statistics: EPC certificates (home features and ratings), NEED (typical energy use) and Land Registry (house prices).

Trick that makes testing meaningful: every fake home also has **hidden traits** (how many people live there, how leaky it really is). These change its bills, but our models never see them, just like real life. Some homes also get a retrofit halfway through, so we can compare our predicted savings with what 'actually' happened.

2. Clean & prepare

Monthly bills become a yearly figure. Gas is mostly used in winter, so 3 summer months cannot simply be multiplied by 4. We weight each month by how much it normally counts. If too many months are missing, we say the figure is unreliable.

3a. The Green Home Score (0-100)

Seven parts, each scored 0-100, then combined with fixed weights set by the problem statement. A better home can never get a lower part-score.

Part	Weight	How we score it (draft)
Energy rating (EPC)	25%	A = 96, B = 86, C = 75, D = 61, E = 46, F = 30, G = 10
Energy use per m^2	20%	50 kWh/m^2 or less = 100, 350 or more = 0, straight line between
Insulation	15%	Loft and walls: none / partial / full
Heating	15%	Heat pump best; modern gas boiler middle; old boiler and direct electric low
Windows	10%	Triple 100, newer double 80, older double 60, partial 35, single 0
Solar / renewables	10%	Share of electricity the panels cover; none = 0
Smart controls	5%	None 0, timer 30, smart thermostat 70, smart + radiator valves 100

80-100	65-79	50-64	30-49	0-29
Highly efficient	Good	Moderate	Poor	Very poor

The dashboard shows how many points each part added, so the customer can see *why*.

3b. 'Expected use' model

A machine-learning model (LightGBM) predicts how much energy a home *like this one* should use, based on size, type, age and region. A big house uses more energy simply because it is big. So we flag a home only when it uses much more than similar homes, which the rules require.

4. Spot poor homes

Homes scoring Poor or Very poor are flagged with their main problems, for example "single glazing" or "old boiler". We rank them by **improvement potential**: the % cut in bought energy (grid electricity + gas) if they did every suitable upgrade.

5. Recommend retrofits

A rulebook decides what is **possible** for each home. Examples: no cavity-wall insulation on solid walls, no new double glazing if it is already double glazed, solar only if there is a usable roof. Each suggestion gets High / Medium / Low suitability and a one-line reason. Order matters: fix the walls and roof first, then controls, then heating, then solar.

6a. Energy savings

We apply upgrades one after another, each acting on what is left. Example (illustrative numbers): loft insulation cuts heating by 15%. The heat pump then only has to cover the smaller remaining demand. Simply adding up each upgrade's saving on its own would overcount.

A heat pump makes gas go down and electricity go up, and we show the electricity increase as a negative saving. Solar lowers what you buy from the grid, not what the home uses.

6b. Carbon

Carbon saved = electricity saved x electricity factor + gas saved x gas factor. Factors come from the UK government's official yearly table (DESNZ). Roughly 0.21 kg CO_2e per kWh for electricity and 0.18 for gas; to be confirmed against the official file.

6c. House value

A model learns house prices from sales data. We ask it the price of the same house, on the same date, before and after the upgrades, and report the difference as a **range** (e.g. +2% to +5%). We always say it is an indicative estimate, not a guarantee.

7-9. Report, AI assistant, dashboard

All results go into one **property report**. The AI assistant can only read that report or ask the code to rerun with different upgrades. A checker then scans every AI answer: any number not in the report is flagged and removed. The dashboard shows the score breakdown, recommendations, savings bars, carbon, value range and a chat box.

Bonus feature: memory

The app remembers each customer, for example "I rent, I can't change the boiler", "budget £5k" or "already did the loft". Later advice adapts to this, while the numbers still come from the code.

Who does what

Persons A and B have stronger AI coding tools, so they take the parts that need a lot of code. Person C takes the parts that need careful thinking, research and checking by hand. Those parts are small in code but worth a lot of marks: the score alone is 15%.

Person A - Data & Machine Learning

Lots of code: data generation and model training.

- Build the fake-town data generator (homes, bills, sales, past retrofits, hidden traits, some missing data on purpose)
- Yearly totals from monthly bills; split data into train / validation / test by customer
- 'Expected use' model and the house-value model, each compared against a simple baseline
- Model test results and final test predictions (submission items G and I)
- Write DATA.md: where data came from, how labels were made, limitations

Person B - Engine, AI assistant & App

Lots of code: wiring everything together, plus the AI and the UI.

- Day 1, hour 1: write the shared data definitions (schema.py) so everyone codes against them
- Savings calculator (electricity and gas, upgrades applied in order) and the 'what if' path
- AI assistant with tools and the number-checker; run the AI tests
- Backend API and the dashboard; customer memory if time allows
- Write LLM.md and draw the architecture diagram (items E and F)

Person C - Scoring, Rules & Docs

Small, careful, easy-to-check pieces. Light AI use is enough.

- Green Home Score: the 7 scoring tables in a config file, the scoring code, tests that weights add to 1 and better homes never score lower
- Retrofit rulebook: when each upgrade is possible, cost range, typical effect, with sources
- Carbon factors from DESNZ and the carbon calculation
- Write ~50 tricky test questions for the AI (missing data, impossible upgrades, heat pump raising electricity)
- Write SCORING.md and the Responsible AI note (items D and H); keep the final checklist

Tip: when Person C needs boilerplate code (file loaders, test scaffolding), A or B can generate it quickly with their tools. C writes the spec, they produce the code.

Timeline & hand-offs

When	What gets handed over	From -> To
Hour 1	Shared data definitions agreed; everyone can start	B -> all
~Hour 4	First rough dataset	A -> B, C
~Hour 6	Retrofit rulebook + carbon factors	C -> B
~Hour 8	Score calculation working	C -> B
~Hour 12	House-value model ready	A -> B
~Hour 14	One home works end to end, from data to dashboard	B
After that	AI tests run on C's questions; results written up	B + C

What we must hand in

Item	Deliverable	Owner
A	Working web app / dashboard	B
B	Code and data scripts, with run instructions	All
C	Trained models and the code that uses them	A
D	Scoring method document	C
E	AI details: prompts, model, how we tested it	B
F	Architecture diagram	B
G	Evaluation report (validation results vs simple baselines)	A + B
H	Responsible AI note	C
I	Held-out test predictions and data documentation	A

Full technical detail (interfaces, file layout, formulas) lives in docs/ARCHITECTURE.md.